



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 415: Communication Engineering

Batch : 22nd
Date: 21/06/2025

Mid-Term Examination

Semester: Spring 2025
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer any five questions:

5 × 6 = 30

1. (a) Is the current Internet using circuit-switching or packet-switching at the data link layer? Explain. 3
(b) Assume we have an isolated link such as a private network in a company. Do we still need addresses in both the network layer and the data-link layer? Explain. 3
2. (a) Given the Dataword 101001111 and the divisor 10111, show the generation of the CRC Codeword at the sender site (using binary division). 3
(b) Create a table for Cyclic Redundancy Check (CRC), where K = 4 and N = 7. 3
3. (a) Draw and explain FSM for the Stop-and-Wait Protocol. 3
(b) State the algorithm to calculate the traditional checksum. 3
4. (a) Explain Control field format for different HDLC frame types. 3
(b) Compare and contrast byte-oriented and bit-oriented protocols. 3
5. (a) Draw a table for Link Control Protocol packet. 3
(b) In PPP, we normally talk about user and system instead of sending and receiving nodes; explain the reason. 3
6. (a) Why is the throughput of Slotted ALOHA 2 times greater than Pure ALOHA? Explain. 3
(b) Sketch the flow diagram for three persistence methods. 3
7. (a) Write the importance of NAV in CSMA/CA. 3
(b) Why is collision an issue in Random access protocol but not in Channelization protocol? 3



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 22nd

Date: 18/06/2025

Mid-Term Examination

Semester: Spring 2025

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following.

5x6= 30

1. (a) Define Artificial Intelligence (AI) 2
(b) Consider a single input neuron defined as: $Y = wx + b$.
Write a python function that returns the neuron's outcome. 3
(c) Which one is not True based on execution speed? 1
a) Interpreter-based AI models take less time.
b) Compiler-based AI model models take less time.
2. (a) Explain AI based on its underlying techniques. 2
(b) A set of student data is given as: 3
 $S = \{ \{ 'r': 1, 'm': 95 \}, \{ 'r': 2, 'm': 80 \}, \{ 'r': 3, 'm': 85 \}, \{ 'r': 4, 'm': 75 \}, \{ 'r': 5, 'm': 60 \} \}$
Write a Python function that returns the average of 'm' (marks).
(c) Is an "if" statement inside another "if" statement possible in Python? 1
3. (a) Explain machine learning. 2
(b) Consider, you are working on an AI project which classifies geometric shapes. One of the features is the area of a given shape. Write a Python function that calculates the area of a triangle, rectangle, and circle. 3
(c) What kind of operator is '()' ? Why is it used? 1
4. (a) Compare Supervised and Unsupervised Learning. 2
(b) Given that a number is defined as, $d = f \times r^e$. Write a Python function that finds the data type of d and explain the reason. 3
(c) Is a tuple a primitive data type in Python? 1
5. (a) Which programming language should you use for real-time AI model deployment? Explain the reason. 2
(b) Consider you are working on a data communication system. Your original data is 5 and 6, but due to a hacker's attack, you are receiving 10 and 12. Explain what is happening there using bitwise operations. 3
(c) What is the equivalent of a switch-case statement in Python? 1
6. (a) Explain deep learning, Statistical Model and Reinforcement learning. 2
(b) What is a control statement? Which iteration statement is a good fit when you only know the stop condition and why? 3
(c) Which programming language has the best eco-system for AI? 1
a) Java b) C++ c) Python d) Prolog e) All of them
7. (a) What is procedural programming? Is it good for AI development? 2
(b) Sigmoid function is defined as: $S = \frac{1}{1+e^{-x}}$. Write a sigmoid function using python. 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE471: Computer and Network Security

Batch: 22nd

Date: 25/06/2025

Mid-Term Examination

Semester: Spring 2025

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

5 × 6 = 30

1. a) State the differences between stream cipher and block cipher. 2
b) Suppose Alice and Bob want to use symmetric key encryption for secure communication. They want to establish their secret key using Diffie-Hellman Key Exchange Algorithm. Show the complete key establishment process with necessary explanation if $g = 19$, $n = 17$, $x = 7$, and $y = 5$. 4
2. a) What is steganography? 1
b) Describe the key management problem in symmetric key encryption. 2
c) What is a digital signature? How will you achieve both authentication and confidentiality using digital signature? Provide necessary illustration and explanation. 3
3. a) Let $p = 7$, $q = 13$ and $e = 5$. Calculate the private and public key using RSA algorithm. 2
b) Suppose message $M = 10$, show the complete encryption and decryption process using the keys you have just calculated. 4
4. a) What is passive attack? 1
b) How can an attacker perform a man-in-the-middle (MITM) attack in an unencrypted communication environment with ensuring the attack remains completely unnoticeable by the authorized users? Explain the process in detail. 2
c) The following text is encrypted using Shift Cipher. Show the cryptanalysis and determine the plaintext. Also determine the key that was used for encryption. 3

SJB DJFW SJB RJ
5. a) State the six key characteristics of Hashing. 2
b) What is a Message Authentication Code (MAC)? Illustrate and explain how a MAC ensures message authenticity and integrity in both encrypted and unencrypted communication. 4
6. a) Explain why 3DES is considered more secure than DES. 2
b) Show the encryption and decryption process of a message 'M' using Elliptic Curve Cryptography (ECC). 4
7. a) Explain why MD5 is considered as a broken hashing algorithm. 2
b) State the limitations of RSA algorithm. 2
c) Show the encryption process of the following message and determine the ciphertext 'C' using Row Column Cipher where row = 4, column = 4 and key = 3124. 2

UNCHARACTERISTIC



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 417: Compiler Design

Date: 28/06/25

Semester: Spring 2025

Mid-Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Answer all questions from the following:

5*6 = 30

- 1 (a) Write down the difference between Compiler and Interpreter. 3
(b) Take an expression for example and show how the Syntax analyzer and the Intermediate code generator work. 3
- 2 (a) Write down the Role of a Parser with diagram. 2
(b) Draw the diagram of the Language processing system. 2
(c) $S \rightarrow AB$
 $A \rightarrow c/aA$
 $B \rightarrow d/bB$
From above grammar draw the parse tree and create a word: **acbd** 2
- 3 (a) What is ambiguous grammar? 2
(b) What is Alphabet and String? Take an example of a string and find the length of the string. 2
(c) Draw a NFA from the Regular Expression: $(a^+ U bba)^+ bab$. 2
- 4 (a) What is DFA? Let $\Sigma = \{a, b\}$ and $L = \{baa\}$, Now design a DFA for L. 3
(b) Convert the following NFA to DFA: $L = \{3^{rd} \text{ symbol from RHS is 'b'}\}$. 3
- 5 (a) $E \rightarrow TE'$
 $E' \rightarrow *TE' \mid \epsilon$
 $T \rightarrow FT'$
 $T' \rightarrow \epsilon \mid +FT'$
 $F \rightarrow id \mid (E$
Find out FIRST() and FOLLOW() from this grammar. 3
(b) $S \rightarrow iCtSS_1 \mid a$
 $S_1 \rightarrow eS \mid \epsilon$
 $C \rightarrow b$
Explain if this grammar will be accept by LL(1) parser or not 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE413: Computer Graphics

Batch: 22nd

Date: 23/06/2025

Mid Term Examination

Semester: Spring 2025

Time: 1 hour and 30 minutes

Total Marks: 30

Answer any five questions from the following:

5x6= 30

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|---|--|---|
| 1 | (a) Define Interactive Computer Graphics. | 2 |
| | (b) What are the driving forces behind computer graphics technology? | 4 |
| 2 | (a) State the differences between Vector and Raster Graphics. | 3 |
| | (b) Explain the basic components of a Cathode Ray Tube (CRT). | 3 |
| 3 | (a) Identify the problems encountered when using raster images. | 2 |
| | (b) Describe the Conceptual Framework for Interactive Graphics and it's key components. | 4 |
| 4 | (a) State the considerations that need to be taken into account when applying basic line drawing algorithms. | 2 |
| | (b) Describe the Digital Differential Analyzer (DDA) line drawing algorithm. | 4 |
| 5 | (a) Outline the principles that guide the Mid-Point Line drawing algorithm. | 2 |
| | (b) Draw a line between the points $P_1(2,3)$ and $P_2(10,8)$ using the DDA line drawing algorithm. | 4 |
| 6 | (a) Define the concept of 8-fold symmetry and determine the pixel-positions for the 45° sector. | 3 |
| | (b) Use Bresenham's algorithm to rasterize the line from $(3, 10)$ to $(10, 4)$. Include each plotted point. | 3 |
| 7 | (a) State the differences between DDA Line Drawing Algorithm and Bresenham's Line Drawing Algorithm. | 3 |
| | (b) Apply the Mid-Point Circle Algorithm to plot a circle with center coordinates $(-3, 5)$ and a radius of 7. | 3 |